COMPUTER ENGINEERING

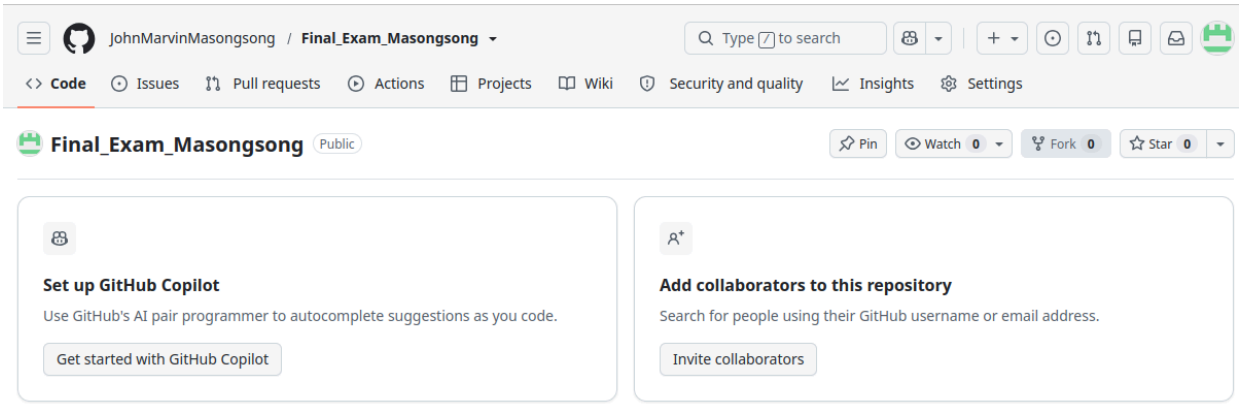
QUEZON CITY

COURSE CODE/TITLE:	CPE212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	10-29-26	**DATE SUBMITTED:**	10-29-26
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ACTIVITY NO. **Final Hands-on Skills Exam**

1. PROCEDURE OUTPUT

1. Create a repository and label it as "Final_Exam_Surname"



Created Repository named “Final_Exam_Masongsong

2. Clone your new repository in your VM

```
masongsong@workstation:~$ ls
ansible-lab        Desktop    Final_Exam_Masongsong  Public     Videos
CPE212_Activity11  Documents  Music                  snap
CPE212_Masongsong  Downloads  Pictures               Templates
masongsong@workstation:~$
```

Cloned “Final_Exam_Masongsong”

3. Create an Ansible playbook that does the following with an input of a config.yaml file and structure inventory file.

Shell

```
---
- name: Install Apache and Node Exporter on all managed hosts
  hosts: all
  become: true
  gather_facts: true

  vars:
    node_exporter_version: "1.12.1"
    node_exporter_arch: "{{ 'amd64' if ansible_architecture == 'x86_64' else
'arm64' }}"

  tasks:
    - name: Install required packages on Ubuntu
      ansible.builtin.apt:
        name:
          - apache2
          - ca-certificates
          - curl
          - tar
        state: present
        update_cache: true
      when: ansible_os_family == "Debian"

    - name: Install required packages on CentOS
      ansible.builtin.dnf:
        name:
          - httpd
          - ca-certificates
          - curl
          - tar
        state: present
      when: ansible_os_family == "RedHat"

    - name: Start and enable Apache on Ubuntu
      ansible.builtin.systemd:
        name: apache2
        state: started
        enabled: true
      when: ansible_os_family == "Debian"

    - name: Start and enable Apache on CentOS
      ansible.builtin.systemd:
        name: httpd
        state: started
        enabled: true
```

```
      when: ansible_os_family == "RedHat"

    - name: Create Node Exporter system group
      ansible.builtin.group:
        name: node_exporter
        system: true
        state: present

    - name: Create Node Exporter system user
      ansible.builtin.user:
        name: node_exporter
        group: node_exporter
        system: true
        shell: /usr/sbin/nologin
        create_home: false
        state: present

    - name: Download Node Exporter
      ansible.builtin.get_url:
        url: "https://github.com/prometheus/node_exporter/releases/download/v{{
node_exporter_version }}/node_exporter-{{ node_exporter_version }}.linux-{{
node_exporter_arch }}.tar.gz"
        dest: "/tmp/node_exporter-{{ node_exporter_version }}.linux-{{
node_exporter_arch }}.tar.gz"
        mode: "0644"

    - name: Extract Node Exporter
      ansible.builtin.unarchive:
        src: "/tmp/node_exporter-{{ node_exporter_version }}.linux-{{
node_exporter_arch }}.tar.gz"
        dest: /tmp
        remote_src: true
        creates: "/tmp/node_exporter-{{ node_exporter_version }}.linux-{{
node_exporter_arch }}/node_exporter"

    - name: Install Node Exporter binary
      ansible.builtin.copy:
        src: "/tmp/node_exporter-{{ node_exporter_version }}.linux-{{
node_exporter_arch }}/node_exporter"
        dest: /usr/local/bin/node_exporter
        remote_src: true
        owner: root
        group: root
        mode: "0755"

    - name: Configure Node Exporter service
      ansible.builtin.copy:
        dest: /etc/systemd/system/node_exporter.service
```

```
    owner: root
    group: root
    mode: "0644"
    content: |
      [Unit]
      Description=Prometheus Node Exporter
      After=network-online.target
      Wants=network-online.target

      [Service]
      User=node_exporter
      Group=node_exporter
      Type=simple
      ExecStart=/usr/local/bin/node_exporter
      Restart=on-failure
      RestartSec=5

      [Install]
      WantedBy=multi-user.target
  notify: Restart Node Exporter

- name: Enable and start Node Exporter
  ansible.builtin.systemd:
    name: node_exporter
    enabled: true
    state: started
    daemon_reload: true

- name: Create Prometheus system group
  ansible.builtin.group:
    name: prometheus
    system: true
    state: present
  when: inventory_hostname in groups['ubuntu']

- name: Create Prometheus system user
  ansible.builtin.user:
    name: prometheus
    group: prometheus
    system: true
    shell: /usr/sbin/nologin
    create_home: false
    state: present
  when: inventory_hostname in groups['ubuntu']

- name: Create Prometheus configuration directory
  ansible.builtin.file:
    path: /etc/prometheus
```

```
        state: directory
        owner: prometheus
        group: prometheus
        mode: "0755"
      when: inventory_hostname in groups['ubuntu']

    - name: Create Prometheus data directory
      ansible.builtin.file:
        path: /var/lib/prometheus
        state: directory
        owner: prometheus
        group: prometheus
        mode: "0755"
      when: inventory_hostname in groups['ubuntu']

    - name: Download Prometheus
      ansible.builtin.get_url:
        url:
"https://github.com/prometheus/prometheus/releases/download/v3.15.1/prometheus-3.
15.1.linux-{{ 'amd64' if ansible_architecture == 'x86_64' else 'arm64' }}.tar.gz"
        dest: "/tmp/prometheus-3.15.1.linux-{{ 'amd64' if ansible_architecture ==
'x86_64' else 'arm64' }}.tar.gz"
        mode: "0644"
      when: inventory_hostname in groups['ubuntu']

    - name: Extract Prometheus
      ansible.builtin.unarchive:
        src: "/tmp/prometheus-3.15.1.linux-{{ 'amd64' if ansible_architecture ==
'x86_64' else 'arm64' }}.tar.gz"
        dest: /tmp
        remote_src: true
        creates: "/tmp/prometheus-3.15.1.linux-{{ 'amd64' if ansible_architecture
== 'x86_64' else 'arm64' }}/prometheus"
      when: inventory_hostname in groups['ubuntu']

    - name: Install Prometheus server binary
      ansible.builtin.copy:
        src: "/tmp/prometheus-3.15.1.linux-{{ 'amd64' if ansible_architecture ==
'x86_64' else 'arm64' }}/prometheus"
        dest: /usr/local/bin/prometheus
        remote_src: true
        owner: root
        group: root
        mode: "0755"
      when: inventory_hostname in groups['ubuntu']
      notify: Restart Prometheus

    - name: Install Prometheus tool binary
```

```
      ansible.builtin.copy:
        src: "/tmp/prometheus-3.15.1.linux-{{ 'amd64' if ansible_architecture ==
'x86_64' else 'arm64' }}/promtool"
        dest: /usr/local/bin/promtool
        remote_src: true
        owner: root
        group: root
        mode: "0755"
      when: inventory_hostname in groups['ubuntu']

    - name: Install Prometheus console files
      ansible.builtin.copy:
        src: "/tmp/prometheus-3.15.1.linux-{{ 'amd64' if ansible_architecture ==
'x86_64' else 'arm64' }}/{{ item }}"
        dest: "/etc/prometheus/{{ item }}"
        remote_src: true
        owner: prometheus
        group: prometheus
        mode: "0755"
      loop:
        - consoles
        - console_libraries
      when: inventory_hostname in groups['ubuntu']

    - name: Configure Prometheus scraping
      ansible.builtin.copy:
        dest: /etc/prometheus/prometheus.yml
        owner: prometheus
        group: prometheus
        mode: "0644"
        content: |
          global:
            scrape_interval: 15s

          scrape_configs:
            - job_name: prometheus
              static_configs:
                - targets: ['localhost:9090']

            - job_name: node_exporter
              static_configs:
                - targets:
                    - '192.168.56.128:9100'
                    - '192.168.56.130:9100'
      when: inventory_hostname in groups['ubuntu']
      notify: Restart Prometheus

    - name: Configure Prometheus systemd service
```

```
    ansible.builtin.copy:
      dest: /etc/systemd/system/prometheus.service
      owner: root
      group: root
      mode: "0644"
      content: |
        [Unit]
        Description=Prometheus Monitoring System
        After=network-online.target
        Wants=network-online.target

        [Service]
        User=prometheus
        Group=prometheus
        Type=simple
        ExecStart=/usr/local/bin/prometheus \
          --config.file=/etc/prometheus/prometheus.yml \
          --storage.tsdb.path=/var/lib/prometheus \
          --web.listen-address=0.0.0.0:9090
        Restart=on-failure
        RestartSec=5

        [Install]
        WantedBy=multi-user.target
    when: inventory_hostname in groups['ubuntu']
    notify: Restart Prometheus

  - name: Enable and start Prometheus
    ansible.builtin.systemd:
      name: prometheus
      enabled: true
      state: started
      daemon_reload: true
    when: inventory_hostname in groups['ubuntu']

  - name: Verify Apache service on Ubuntu
    ansible.builtin.command:
      cmd: systemctl is-active apache2
    changed_when: false
    when: ansible_os_family == "Debian"

  - name: Verify Apache service on CentOS
    ansible.builtin.command:
      cmd: systemctl is-active httpd
    changed_when: false
    when: ansible_os_family == "RedHat"

  - name: Verify Node Exporter service
```

```
    ansible.builtin.command:
      cmd: systemctl is-active node_exporter
    changed_when: false

  - name: Verify Prometheus service
    ansible.builtin.command:
      cmd: systemctl is-active prometheus
    changed_when: false
    when: inventory_hostname in groups['ubuntu']

handlers:
  - name: Restart Node Exporter
    ansible.builtin.systemd:
      name: node_exporter
      state: restarted
      daemon_reload: true

  - name: Restart Prometheus
    ansible.builtin.systemd:
      name: prometheus
      state: restarted
      daemon_reload: true
    when: inventory_hostname in groups['ubuntu']
```

3.1 Install and configure one enterprise service that can be installed in Debian and Centos servers

```
masongsong@workstation:~/Final_Exam_Masongsong$ ansible all -i inventory.ini -b
-m shell -a "systemctl is-active apache2 || systemctl is-active httpd" -K
BECOME password:
192.168.56.128 | CHANGED | rc=0 >>
active
192.168.56.130 | CHANGED | rc=0 >>
inactive
active
masongsong@workstation:~/Final_Exam_Masongsong$
```

3.2 Install and configure one monitoring tool that can be installed in Debian and Centos servers (if it is a stack there should be option of different host)

```
masongsong@workstation:~/Final_Exam_Masongsong$ ansible ubuntu -i inventory.ini
-b -m command -a "systemctl status prometheus --no-pager" -K
BECOME password:
192.168.56.128 | FAILED | rc=4 >>
Unit prometheus.service could not be found.non-zero return code
```

4.4 Change Motd as "Ansible Managed by <username>"

```
masongsong@server1:~$ cat /etc/motd
Welcome to server1.
This was manageed by ansible. Unauthorized access is prohibited.
masongsong@server1:~$
```

```
Last login: Thu Oct 29 10:26:37 2026 from 192.168.56.127
cat[masongsong@centos ~]$ cat /etc/motd
Welcome to centos.
This was manageed by ansible. Unauthorized access is prohibited.
[masongsong@centos ~]$
```

Output of Configuring MOTD

4. Push and commit your files in GitHub

https://github.com/JohnMarvinMasongsong/Final_Exam_Masongsong

5. Make sure to show evidence of input (codes) process (codes successfully running) and output (evidence of installation)

5. For your final exam to be counted, please paste your repository link as an answer in this exam.

Note: Extra points if you will implement the said services via containerization.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

https://github.com/JohnMarvinMasongsong/Final_Exam_Masongsong

COMPUTER ENGINEERING

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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